Adventist University of Central Africa
P.O. Box 2461 Kigali, Rwanda | www.auca.ac.rw | info@auca.ac.rw

Faculty of Information Technology
Capstone Project Academic Year: 2024-2025
Course Code & Name: Database Development with PL/SQL INSY 8311
Lecturer: Eric Maniraguha | eric.maniraguha@auca.ac.rw Date: March 21, 2025
All Groups (Mon, Tues, Wed, Thu)

Instructions: The project will be completed in phases, each with specific requirements. Please keep the following points in mind:

1. **Respect Deadlines:**
    - Deadlines are strict. Late submissions will not be accepted and will result in a score of zero for that phase.
    - Plan and manage your time effectively to ensure timely submissions.
2. **Proposal Submission:**
    - This is a personal project. Late submissions of the proposal will not be accepted and will result in zero points.
3. **Active Participation:**
    - You are expected to fully engage with the project and understand every aspect of it.
    - Ensure you are actively working and contributing to each phase.
4. **Focus on Quality:**
    - Aim for high-quality work in every aspect of the project. Quality will be a key factor in project acceptance.
    - Review your work carefully before submission to avoid last-minute delays.
5. **Exploration of Ideas:**
    - Feel free to explore different ideas during the implementation phase.
    - You may submit multiple ideas, and I will assist you in selecting the best one in the first phase.
6. **Project Acceptance:**
    - Not all projects will be accepted. If your proposal does not meet the required standards, you will be given two additional days to refine or rewrite it.
    - Make sure your proposal is well thought out and clearly defined to avoid this situation.
7. **Report Submission via GitHub:**
    - After the problem statement, you must submit the report for each subsequent phase via GitHub.
    - I will be monitoring all submissions through GitHub, so make sure to keep it updated regularly.

Note: This project is considered your final exam.

Project Outline: PL/SQL Oracle Database Design and Implementation

This project is a comprehensive, multi-phase individual assignment focused on Oracle database design and PL/SQL programming. You will select a complex, real-world problem and develop a working database solution through multiple clearly defined phases.

I. Phase: Problem Statement and Presentation

- **Deadline: March 25, 2025**, at **11:59 PM**

1. Objective

Identify a real-world problem that requires a PL/SQL-based Oracle database solution. The problem should be complex enough to involve **multiple entities, relationships, and business logic**.

2. Problem Statement Guidelines

Your submission should clearly outline:

- **Problem Definition:** What issue are you solving?
- **Context:** Where and how will the system be used? (e.g., hospitals, universities, businesses)
- **Target Users:** Who will benefit from your system?
- **Project Goals:** What outcomes are expected? (e.g., automate workflows, improve accuracy, enhance data security)

3. Presentation Requirements

Prepare a brief in-class presentation that explains:

- The **project objectives**
- The **main entities** in your database
- The **anticipated benefits** of your solution

4. Document Format

Please submit your work as a **PowerPoint presentation** with the following formatting:

- **Font:** Helvetica
- **Use bullet points** throughout the slides
- **Number of slides:** Maximum of **three slides** (you may submit one or two slides only)
- Ensure clarity and consistency in layout

5. File Naming Convention

- Name your file using the following format:
- **Grp_studentID_firstName_PLSQL**
- *Example:* Mon_121212_eric_plsql
- Make sure to use **underscores** (_) between each part.

II. Phase: Business Process Modeling (Related to Management Information Systems - MIS)

Objective

In this phase, you will model a business process relevant to Management Information Systems (MIS). The goal is to visualize how information flows within a system, how different entities interact, and how decision-making is supported through MIS.

Project Requirements

1. **Define the Scope**
   - Clearly outline the business process you are modeling.
   - Ensure the process is relevant to MIS, such as **order processing, inventory management, patient record handling, employee payroll, or customer service management**.
   - Define the **objectives and expected outcomes** of the process.

2. **Identify Key Entities**
   - List and describe all key **actors, departments, or systems** involved in the process.
   - Explain their **roles, responsibilities, and interactions** within the system.
   - Examples of entities: **users (employees/customers), databases, applications, managers, automated workflows**.

3. **Use Swimlanes for Clarity**
   - Organize your diagram using **swimlanes** to separate different actors or departments.
   - Ensure the swimlanes make it easy to understand who is responsible for each step of the process.

4. **Apply UML/BPMN Notations**
   - Use **UML (Unified Modeling Language) or BPMN (Business Process Model and Notation)** to represent your process visually.
   - Maintain consistency with proper symbols and conventions.
   - Include key elements such as **start and end points, tasks, decisions, and data flows**.

5. **Ensure a Logical Flow**
   - The process model should clearly **flow from start to end** with all interactions properly mapped.
   - Highlight **decision points, inputs, outputs, and system interactions**.
   - Ensure that **dependencies between steps** are correctly structured.

6. **Prepare an Explanation**
   - Write a brief **description of your diagram** (one page maximum).
   - Explain:
     - **The main components and their interactions.**
     - **How the process supports MIS functions** (e.g., improves decision-making, streamlines operations).
     - **Why this process is important for organizational efficiency.**
     - Tools [Lucichart] or [draw.io]

Additional Notes

- If you have **not taken the MIS course**, research how **information systems support business processes** to ensure your model aligns with MIS principles.
- Exceptional work may receive **bonus points** for **clarity, creativity, and technical accuracy**.
- This phase builds the foundation for later stages of database design, so ensure accuracy and completeness.

III. Phase: Logical Model Design

Objective:
In this phase, you will design a **detailed logical data model** for your project, ensuring it aligns with the problem statement defined in **Phase 1** and the business process modeled in **Phase 2**.

Requirements:

1. **Entity-Relationship (ER) Model:**
   - Identify and define all **entities** relevant to your project.
   - Specify **attributes** for each entity, including data types.
   - Clearly define **primary keys (PKs)** and **foreign keys (FKs)** to establish relationships.

2. **Relationships & Constraints:**
   - Establish and document all **relationships** between entities (e.g., one-to-one, one-to-many, many-to-many).
   - Apply **constraints** such as **NOT NULL**, **UNIQUE**, **CHECK**, and **DEFAULT** where necessary.

3. **Normalization:**
   - Ensure that your database design **eliminates redundancy** and adheres to normalization principles (**at least 3rd Normal Form - 3NF**).

4. **Handling Data Scenarios:**
    - Validate that your logical model can handle different **real-world data scenarios** effectively.

5. **Presentation & Feedback:**
    - Submit your logical model for review and feedback before proceeding to the next phase.
    - Your model should be **well-documented** and **clearly labeled** for readability.

Submission Method

- **Phase I (Project Description/Problem Statement), Phase II (Business Process Modeling), and Phase III (Logical Model Design)** must be combined into a **single, concise report** and submitted via GitHub.
- Create a **private GitHub repository** that includes **all phases**, ensuring the report is **clear and concise**.
- The report should include:
    - **Screenshots** of your **Business Process Model (BPM)** and **Conceptual Model**.
    - **Short descriptions** explaining each diagram.

Repository Naming Format:

- **Grp_StudentID_ProjectName**
    - **Example:** Mon_121212_StudentMS, Tues_121212_StudentMS, etc.
- **Access & Submission:**
    - **Add me as a collaborator** (*eric.maniraguha@auca.ac.rw*) to grant access.

Important Notes:

- The repository **must start with your name and Student ID** for identification.
- **Late submissions will not be accepted.**

Additional Notes

- **Independent Work:** Each student is expected to work independently and complete all project phases step by step.
- **Creativity & Complexity:** Choose a unique, meaningful problem. Projects that demonstrate originality and thoughtful design will be rewarded.
- **Progress Tracking:** Regular updates and presentations will be required to assess your progress.
- **Use of AI Tools:** AI tools may be used for research and paraphrasing. However, **plagiarism will not be tolerated**. Ensure all content is your original work.
- **Bonus Opportunity:** Exceptional performance during each phase may earn you **2 bonus points**.

- **Upcoming Phases:** Additional project phases will be shared as the course progresses.
- **No Excuses:** Deadlines are strict. No delays will be tolerated unless you're repeating the course. Stay disciplined and focused throughout.

Final Note:
This project serves as your **Final Exam**. Give it your best effort! Your commitment and creativity will define your success.

For I know the plans I have for you," declares the Lord, "plans to prosper you and not to harm you, plans to give you hope and a future."
— Jeremiah 29:11 (NIV)

Good luck!